Complete Baseline Evidence: Supplement to MRI Interpolation Quality Assessment

1 Complete Baseline Ledger and Training-Budget Sensitivity

The experiment master table contains 13 reconstruction baselines. We recover all 39 three-cohort $\times 4$ rows, all 52 fastMRI/NingXia $\times 2/\times 6$ rows, and all 20 recorded longer-budget rows: 111 configurations and 555 metric values. Original metric summaries agree to their reported four-decimal precision. Two overwritten SAINT summaries at fastMRI $\times 2/\times 6$ are recovered from all 48 preserved output volumes. No run is retrained for this audit.

The native evaluator measures every nonacquired plane on the whole image, averages planes within an examination, and then averages examinations. NingXia first averages series within a held-out patient fold, then averages all eight folds. PSNR uses each target plane's range; SSIM is the full-plane implementation. These are reference-based evaluation metrics, not deployable QA scores. Each configuration has matching case identities and generated-plane counts across its models; this count audit does not independently certify all historical preprocessing. The old masked common-support tables and this native-metric ledger remain distinct contracts. Automatic threshold-mask Dice/HD95 columns are not interpreted as anatomical segmentation.

All model names denote the local implementations, including generic SR models adapted to 2.5-D conditioning, inherited GAN, and local SAINT/CycleINR variants. The recorded master profiles include different training budgets and early stopping; no equal-compute claim is made. PAST-Net's fastMRI development trajectory remains development evidence rather than a new independent three-cohort baseline result.

What the full bank adds. Using all 13 native-metric baselines gives 78 model pairs per cohort. Median case-error rank correlations are .997 on IXI, .899 on fastMRI, and 1.000 on eight NingXia patient folds. These descriptive estimates show shared difficulty across the full bank while retaining the lower fastMRI association than in the masked analysis. They do not imply independent model pairs or extend the five-model intervention to 13 models.

Budget protocol and negative boundaries. For every recorded 60/90-epoch experiment, compare its matched standard-profile cases; use 10,000 paired bootstrap draws (seed 20260910), examinations for fastMRI and patient folds for NingXia. The primary MAE intervals simultaneously cover all 20 comparisons using the bootstrap maximum standardized deviation. All five metric changes remain in the numerical record. Fifteen MAE intervals favor longer training; five cross zero. Runs are separate training experiments, not a controlled single-checkpoint continuation, and standard values are never replaced by their best budget.

NingXia Res-SRDiff $\times 2$ MAE changes from 440.61 at 30 epochs to 140.45 at 60 and 118.25 at 90, still above Linear's 78.53. A short-budget failure therefore cannot establish an intrinsic diffusion-family failure. Conversely, CycleINR $\times 2$ barely changes (.000006 MAE reduction), and five simultaneous intervals remain inconclusive. The source note that fastMRI ACVTT $\times 6$ worsened at 60 epochs is contradicted by its numbers: PSNR improves 19.8309 to 20.0269, and MAE 72.4367 to 70.9050; all five metric directions improve. The original note is preserved and this correction is explicit.

Table 1: IXI ×4: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	26.1722	0.8529	29.2781	6553.8673	0.9390
Cubic	25.7138	0.8422	31.1269	7217.2979	0.9303
ACVTT	27.3431	0.8822	25.5217	5206.8325	0.9557
Res-SRDiff	28.0441	0.8821	24.1938	4664.4129	0.9611
ArSSR	27.2467	0.8773	25.5509	5267.3658	0.9533
CycleINR-style	26.1728	0.8527	29.2765	6553.8469	0.9390
ResUNet	27.4611	0.8812	25.2395	4985.4989	0.9552
Pix2Pix-GAN	27.1070	0.8506	27.0202	5372.2668	0.9538
SwinIR	28.6485	0.8997	22.4556	4137.5109	0.9658
SAINT-style	28.6677	0.8970	22.8365	4128.0151	0.9660
I3Net	31.3058	0.9285	17.5276	2565.3177	0.9788
HAT	28.7249	0.9016	22.2500	4088.7422	0.9663
SMamba-UNet	29.3536	0.9077	21.1352	3691.2381	0.9699

Table 2: fastMRI ×4: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	19.1133	0.5970	78.8111	23092.0046	0.7733
Cubic	17.9736	0.5332	91.9330	29740.2642	0.7285
ACVTT	20.1941	0.6624	71.5480	18286.6190	0.8320
Res-SRDiff	21.6727	0.6909	56.3371	13095.4191	0.8814
ArSSR	20.3143	0.6575	65.9931	17745.8948	0.8301
CycleINR-style	20.3156	0.6577	66.0985	17743.2358	0.8302
ResUNet	22.2549	0.7374	53.5705	11842.8147	0.8955
Pix2Pix-GAN	21.9090	0.7294	53.2622	12485.1668	0.8915
SwinIR	22.3578	0.7494	50.8733	11636.3791	0.8977
SAINT-style	23.2605	0.7705	46.4417	9347.6662	0.9177
I3Net	24.0102	0.7921	41.0101	7755.6546	0.9308
HAT	23.0894	0.7688	47.2467	9773.1765	0.9138
SMamba-UNet	22.7322	0.7618	49.4487	10826.4468	0.9051

Table 3: NingXia ×4: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	22.7335	0.6749	103.2052	42189.1127	0.8576
Cubic	21.6460	0.6184	119.4078	53313.5368	0.8252
ACVTT	23.4735	0.7281	95.4376	35779.4840	0.8804
Res-SRDiff	21.1538	0.5538	153.0213	66533.1526	0.8207
ArSSR	23.1634	0.7091	96.4890	38343.9409	0.8708
CycleINR-style	22.9009	0.6935	101.0990	40735.3568	0.8625
ResUNet	23.3816	0.7209	97.0409	36316.2290	0.8785
Pix2Pix-GAN	22.4530	0.7112	105.8119	46398.6707	0.8552
SwinIR	23.9513	0.7525	85.9679	32030.5455	0.8930
SAINT-style	23.8350	0.7422	89.4140	32992.9496	0.8905
I3Net	23.4911	0.7208	93.5655	35725.4805	0.8825
HAT	23.9499	0.7519	86.1449	32226.9860	0.8933
SMamba-UNet	23.9126	0.7500	86.8152	32498.3175	0.8917

Table 4: fastMRI ×2: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	21.6348	0.7030	53.9754	11591.9388	0.8732
Cubic	20.3321	0.6363	64.1081	15236.6373	0.8370
ACVTT	21.7910	0.7110	53.4000	11160.0709	0.8783
Res-SRDiff	22.4766	0.6511	50.3378	9901.7857	0.9034
ArSSR	23.1137	0.7613	43.5812	8391.3334	0.9140
CycleINR-style	23.0957	0.7593	43.7063	8411.5115	0.9132
ResUNet	24.1700	0.7913	39.7905	6909.6572	0.9338
Pix2Pix-GAN	26.0734	0.8302	31.2304	4584.5238	0.9565
SwinIR	24.8716	0.8153	35.9392	5990.3711	0.9428
SAINT-style	24.4308	0.7809	39.6861	6749.7212	0.9373
I3Net	27.7711	0.8670	25.3325	3113.5377	0.9700
HAT	24.6475	0.8106	37.2729	6225.7429	0.9409
SMamba-UNet	24.6548	0.8077	37.1110	6361.0863	0.9401

Table 5: fastMRI ×6: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	18.2648	0.5595	88.2647	28258.4309	0.7215
Cubic	18.2648	0.5595	88.2647	28258.4309	0.7215
ACVTT	19.8309	0.6601	72.4367	20031.3896	0.8120
Res-SRDiff	21.2529	0.6791	59.4388	14747.2531	0.8702
ArSSR	19.4214	0.6156	74.9270	21914.5364	0.7872
CycleINR-style	19.3873	0.6158	74.9017	22086.6021	0.7854
ResUNet	22.1434	0.7375	53.8609	12277.0333	0.8921
Pix2Pix-GAN	21.6510	0.7250	54.8359	13567.5373	0.8828
SwinIR	21.8398	0.7305	56.4490	13267.8010	0.8840
SAINT-style	22.3172	0.7466	52.5303	11683.3596	0.8979
I3Net	22.8308	0.7617	47.0732	10309.8900	0.9088
HAT	21.8230	0.7313	55.7775	13341.4608	0.8829
SMamba-UNet	21.9153	0.7313	56.5002	13166.6662	0.8848

Table 6: NingXia ×2: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	24.9741	0.7264	78.5349	27551.4768	0.9112
Cubic	24.1081	0.6767	89.5215	33708.1987	0.8952
ACVTT	25.3139	0.7518	76.2396	25631.8031	0.9180
Res-SRDiff	15.3187	0.2945	440.6059	311894.3839	0.8320
ArSSR	25.1538	0.7427	76.6719	26571.4634	0.9146
CycleINR-style	25.1538	0.7427	76.6723	26571.4525	0.9146
ResUNet	25.2629	0.7478	76.6937	25684.3115	0.9170
Pix2Pix-GAN	25.2116	0.7543	74.8168	25448.9329	0.9169
SwinIR	25.5845	0.7651	73.1350	24249.0709	0.9226
SAINT-style	25.3398	0.7535	75.9435	25247.7430	0.9191
I3Net	25.7679	0.7574	70.8835	23483.9767	0.9265
HAT	25.8489	0.7738	70.0441	22767.1915	0.9276
SMamba-UNet	25.4409	0.7603	74.6767	24857.0321	0.9201

Table 7: NingXia ×6: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	21.7375	0.6509	118.6269	53812.4881	0.8146
Cubic	21.6523	0.6483	119.6982	54616.6079	0.8118
ACVTT	22.6160	0.7126	108.2221	44622.4723	0.8490
Res-SRDiff	20.4586	0.5616	156.7453	75246.6372	0.7890
ArSSR	22.3530	0.7007	108.1897	47147.2527	0.8389
CycleINR-style	21.8961	0.6728	116.1079	52220.1671	0.8214
ResUNet	22.7683	0.7143	104.7535	42861.1183	0.8551
Pix2Pix-GAN	21.1353	0.6662	129.1480	61058.8712	0.8005
SwinIR	22.9792	0.7362	98.5621	40916.0806	0.8624
SAINT-style	22.7341	0.7233	106.0398	44729.0092	0.8562
I3Net	22.3951	0.6940	111.4996	46546.2263	0.8452
HAT	23.0110	0.7361	98.5351	40588.4310	0.8635
SMamba-UNet	22.9507	0.7336	100.4715	41414.3824	0.8611

Table 8: All recorded extended-budget results. Native five-metric values are kept separately from the master-profile ranking.

Cohort	Method	Factor	Epochs	PSNR	SSIM	MAE	MSE	NCC
NX	ArSSR	4	60	23.4566	0.7241	91.8598	36115.4376	0.8793
NX	ResUNet	4	60	23.7562	0.7379	90.5450	33771.8800	0.8876
NX	SAINT-style	4	60	23.8388	0.7473	88.1769	33105.9966	0.8909
NX	SMamba-UNet	4	60	24.0160	0.7542	84.8084	31545.2244	0.8947
NX	SwinIR	4	60	24.0370	0.7573	83.8378	31317.4826	0.8946
NX	CycleINR-style	2	60	25.1538	0.7427	76.6723	26571.4690	0.9146
NX	HAT	2	60	26.1387	0.7853	66.2919	21520.2802	0.9321
NX	Res-SRDiff	2	60	22.6714	0.5619	140.4503	56206.7760	0.8894
NX	SAINT-style	2	60	25.6706	0.7677	72.4046	23699.0011	0.9245
NX	SMamba-UNet	2	60	25.9101	0.7813	68.4655	22757.3826	0.9281
NX	SwinIR	2	60	26.0556	0.7821	67.3308	21930.8546	0.9310
NX	ArSSR	6	60	22.4720	0.7020	106.4477	46384.2568	0.8423
NX	CycleINR-style	6	60	22.1578	0.6886	111.6263	49569.1804	0.8335
NX	Res-SRDiff	6	60	20.8328	0.6008	139.9794	62636.9837	0.7782
NX	ResUNet	6	60	22.9161	0.7281	100.4640	41591.4635	0.8610
NX	SwinIR	6	60	23.0201	0.7392	97.7710	40687.7343	0.8637
NX	Res-SRDiff	2	90	23.5044	0.6178	118.2457	43546.3681	0.8924
fastMRI	SwinIR	2	60	25.4180	0.8274	33.7890	5326.9072	0.9500
fastMRI	ACVTT	6	60	20.0269	0.6677	70.9050	19173.6891	0.8211
fastMRI	Res-SRDiff	2	60	24.0661	0.7540	40.8709	7086.3766	0.9320

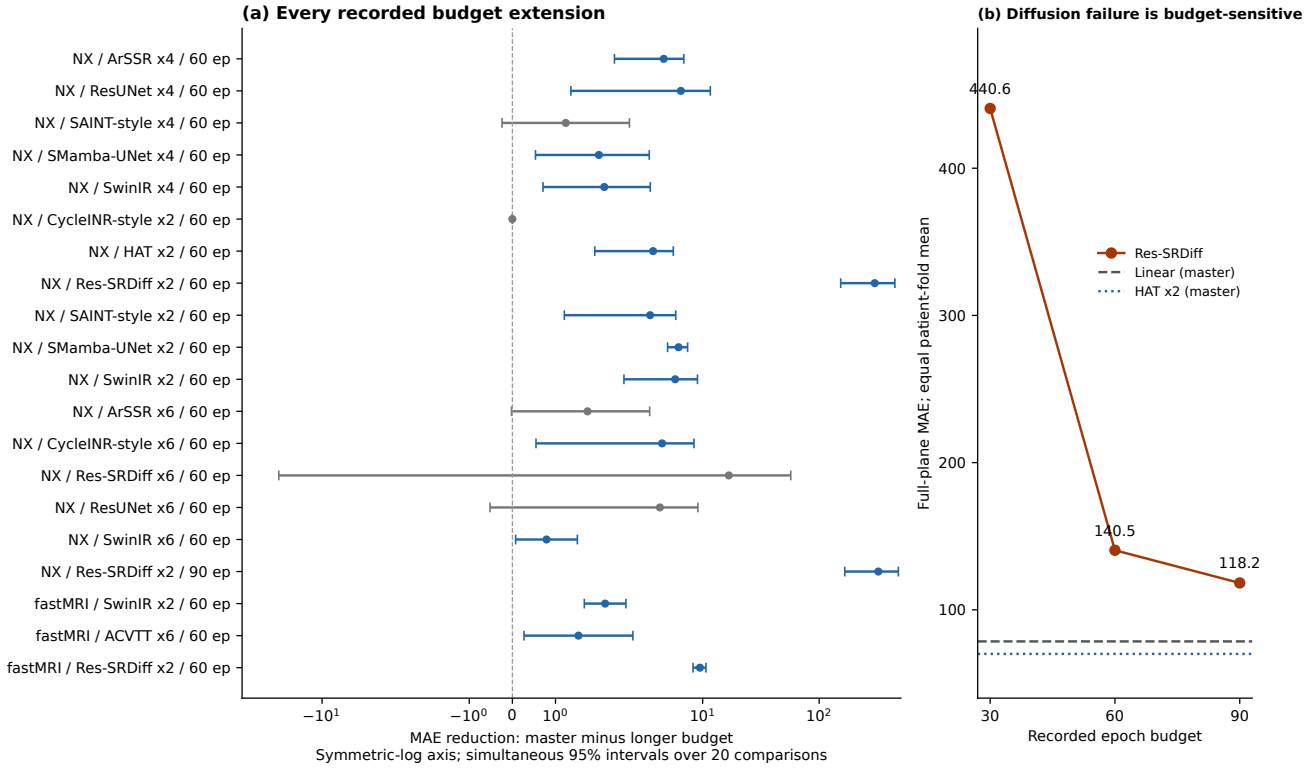


Figure 1: All 20 recorded budget extensions. Positive reductions favor longer training; the symmetric-log axis keeps the large diffusion changes and small effects visible. Intervals simultaneously cover the 20 primary MAE contrasts. Five cross zero. Budget sensitivity does not validate a new QA score or erase the original short-budget failure.